Create the Experimental Design in a .txt file

Below is the main information to include in the experimental design .txt file:

The file should contain **4 columns** with the following names:

"Name", "Experiment", "Groups", and **"Type"**

Each column represents the following information:

- Name: The name of the measured sample as it appears in the input file with the measurements (except for MaxQuant files, where it represents the name you would like to assign to each condition). More details can be found below.
- Experiment: This column contains a unique name for each measured sample.
- Groups: The name of the biological condition for each sample.
- Type: Defines which samples **will be considered in the analysis ("T"), which will not (empty cell),** and **which are controls ("C"),** by using T, C, or leaving the cell empty.

Note:

You don't need to follow the same column order as in the input file. You can list the entries in the order you'd like them to appear in the output (plots and tables). SafeQuant will follow the order specified in the experimental design for all measurements—except for the control, which will be placed first.

Additional columns with extra information can be included in the file, but they will not be used by SafeQuant.

Experimental Design Format for Progenesis Files

If the Progenesis file has the format shown in the picture below:

AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE
Raw abundance													
A			B			C			D			E	
Measure_1	Measure_2	Measure_3	Measure_4	Measure_5	Measure_6	Measure_7	Measure_8	Measure_9	Measure_10	Measure_11	Measure_12	Measure_13	Measure_14

Then, the experimental design should have the following format:

Name	Experiment	Groups	File
Measure_1	Sample_1	Control	C
Measure_2	Sample_2	Control	C
Measure_3	Sample_3	Control	C
Measure_4	Sample_4	Condition_1	T
Measure_5	Sample_5	Condition_1	T
Measure_6	Sample_6	Condition_1	T
Measure_7	Sample_7	Condition_2	T
Measure_8	Sample_8	Condition_2	T
Measure_9	Sample_9	Condition_2	T
Measure_10	Sample_10	Condition_3	T
Measure_11	Sample_11	Condition_3	T
Measure_12	Sample_12	Condition_3	T
Measure_13	Sample_13	Condition_4	T
Measure_14	Sample_14	Condition_4	T

<u>Experimental Design Format for MaxQuant files:</u>

If the MaxQuant file has the name of the columns as shown in the picture below:

CL	CM	CN
LFQ intensity AD4553 Secreting medium-2	LFQ intensity AD4553 in Non-Secreting medium-1	LFQ intensity AD4553 Non-Secreting medium-2

Then, the experimental design should have the following format:

	A	B	C	D
1	Name	Experiment	Groups	File
2	Measurement 1	AD4553 Secreting medium-1	Condition 2	T
3	Measurement 2	AD4553 Secreting medium-2	Condition 2	T
4	Measurement 3	AD4553 Secreting medium-3	Condition 2	T
5	Measurement 4	AD4553 in Non-Secreting medium-1	Condition 1	T
6	Measurement 5	AD4553 Non-Secreting medium-2	Condition 1	T
7	Measurement 6	AD4553 Non-Secreting medium-3	Condition 1	T
8	Measurement 7	WT + pSW039 Secreting medium-1	Control	C
9	Measurement 8	WT + pSW039 Secreting medium-2	Control	C
10	Measurement 9	WT + pSW039 Secreting medium-3	Control	C